

The study of uniform convergence in the representation of functions using Computer Algebra Systems

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The representation of functions through sequences and functional series is a topic of utmost importance within Mathematical Analysis [1]. The concept of uniform convergence, whose understanding has historically been a difficult obstacle for most students to overcome, occupies a central place within this problem. In this work, some of the activities carried out in the subject are presented, discussed and analysed, with the support of CAS, to reconnect the different representations [2] and achieve a greater understanding of the fundamental concepts and results linked to this topic.

The discussion focuses on the influence of the types of convergence on the properties of the limit functions of sequences or sum of series. Likewise, tasks linked to the different uniform convergence criteria are proposed to achieve understanding of them through graphic visualization [3], for example, of the supreme criterion, graphically appreciating the tendency towards zero of its value; or the use of a major numerical sequence to demonstrate uniform convergence. There were also cases in which the visualization seemed to reflect nothing and by suggesting scale changes in some of the axes, the expected behaviours could be seen.

Keywords

uniform convergence, visualization, representation of functions

References

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